Infinex 2k26

INFINEX 2K26 — GLOBAL AI HACKATHON • TRACK: AGENTIC AI & AUTOMATION

AI-Powered Disaster Message Prioritization & Response Automation

RescueFlow AI is an AI-powered emergency intelligence and disaster-response coordination platform that transforms unstructured citizen distress messages into structured, prioritized and actionable emergency incidents.

CENTRAL INNOVATION: AI UNDERSTANDS → n8n AUTOMATES → MONGODB REMEMBERS → REACT VISUALIZES → HUMAN APPROVES

AGENTIC AI & INFERENCE

Groq Llama 3.3-70B

Sub-2s disaster classification & entity extraction with strict JSON schema.

WORKFLOW ORCHESTRATION

n8n Automation Engine

Central event bus linking incoming triggers, validation, and multi-channel dispatches.

SAFETY & DETERMINISM

Deterministic Scoring

Pure mathematical 8-factor life-safety score (0–100) preventing AI hallucinations.

HUMAN COMMAND CONTROL

Human-in-the-Loop Gate

Operations Commander sign-off required before activating autonomous field dispatches.

Five Critical Failures in Disaster Operations

1. Disaster Communication Overload: During floods, fires, and cyclones, emergency control rooms receive massive surges of unstructured citizen reports simultaneously across phone lines, messaging apps, and email.

2. Manual Emergency Triage: Human dispatchers must manually read, interpret, categorize, and log messages into CAD screens, creating catastrophic processing delays during crisis peaks.

3. Duplicate Incident Reports: Dozens of citizens repeatedly report the identical emergency. Without intelligent correlation, duplicate records scatter scarce ambulances and rescue boats.

4. Critical Emergencies Get Buried: High-consequence reports involving trapped victims, medical trauma, infants, or elderly citizens get lost among routine questions and non-urgent messages.

5. Unreliable AI-Only Decisions: Unconstrained, purely generative LLMs are prone to hallucinations, unsupported priority assignments, and unpredictable failures in life-or-death situations.

Operational Consequences

- Delayed identification of critical life-or-death emergencies
- Inefficient resource allocation & duplicate deployments
- Fragmented, redundant records across isolated departments
- Severely restricted real-time situational awareness
- Extreme dispatcher burnout and cognitive overload

CORE IMPERATIVE:

“Emergency response needs more than message collection. It needs intelligent triage, coordinated automation and accountable human decisions.”

END-TO-END PROCESSING PIPELINE:

CITIZEN REPORT → MULTI-CHANNEL INTAKE → FASTAPI/n8n ORCHESTRATION → GROQ LLAMA 3.3-70B AI → STRUCTURED EXTRACTION → DEDUPLICATION → DETERMINISTIC PRIORITY → MONGODB → REACT COMMAND CENTER + GIS → HUMAN APPROVAL → n8n DISPATCH → TELEGRAM/EMAIL ALERTS

1. INTAKE & AI EXTRACTION

FastAPI, Groq Llama 3.3-70B & Heuristic Fallback

- Ingests multi-channel reports from Telegram, web forms, emails, and CSV batch files into canonical JSON.
- Groq Cloud API running Llama 3.3-70B Versatile classifies active emergency validity in ~1.1 seconds.
- Extracts structured crisis parameters: hazard type, trapped victims, vulnerable groups, and tactical gear.
- Automatically filters non-emergency casual chats to prevent dispatcher distraction.

2. DEDUPLICATION & SCORING

Semantic Deduplication & Deterministic Prioritization

- Deduplication engine evaluates token Jaccard similarity (threshold >= 0.75) and spatial landmark recognition.
- Consolidates corroborating citizen reports into a single master incident with merged timeline history.
- Peak victim count damping prevents accidental inflation of casualties across repeated calls.
- Pure deterministic engine calculates 0–100 priority score across 8 transparent safety criteria with zero hallucinations.

3. n8n AUTOMATION & HUMAN GATE

n8n Master Orchestrator & Commander Approval

- n8n is the central workflow automation and orchestration layer connecting intake, triage, approval, and notifications.
- Mandatory Human-in-the-Loop gate requires Operations Commander authorization before critical dispatches occur.
- Approved incidents trigger n8n visual workflows to dynamically route dispatches to Telegram responder channels & emails.
- Real-time Server-Sent Events push live state to Leaflet GIS map with pulsing markers in <500ms.

Infinex 2k26

🌐 Multi-Channel Emergency Intake

Interactive web reporting portal, continuous asynchronous Telegram bot polling (@RescueFlowAI_DemoBot), configured email triggers, and multi-row CSV batch imports.

🛡️ Human-in-the-Loop Approval Gate

Dedicated review interface where authorized coordinators examine extracted intelligence, GIS coordinates, and approve or reject P1/P2 dispatches with recorded reasons.

⚡ Groq Llama 3.3-70B AI Intelligence

Sub-1.5s emergency intent classification, disaster parameter extraction, structured JSON generation, automatic self-repair retry, and zero-crash heuristic regex fallback.

🗺️ Real-Time GIS Command Center

Interactive Leaflet OpenStreetMap with pulsing severity markers, 4-tier geocoding (GPS, regex, 70+ Coimbatore gazetteer, sector fallback), and sub-second SSE streaming.

🔄 Semantic Incident Deduplication

Jaccard token similarity (>= 0.75) and landmark proximity engine to consolidate corroborating reports, eliminate duplicate tickets, and record full report history.

📢 Automated Emergency Notifications

Autonomous n8n workflow execution delivering dynamic markdown alerts to first responder Telegram channels, configured email queues, and immutable audit logs.

🎯 Deterministic Priority Engine

Transparent, explainable 0–100 priority scoring based on 8 safety factors (immediate life danger, trapped victims, vulnerable groups, scale), mapped to P1–P4 tiers.

ENGINEERED FOR CRISIS RELIABILITY:

RescueFlow AI delivers rapid AI triage without sacrificing operational safety. Every physical dispatch order remains accountable, transparent, and strictly supervised by human commanders.

Infinex 2k26

💻 Frontend: React 18 & Tactical Console

- React 18, Vite, React Router, Axios, React-Leaflet, OpenStreetMap
- Provides EOC command center, live incident monitoring, approval UI, and GIS map.

⚙️ Backend: FastAPI & Python 3.13

- Python 3.13, FastAPI (Async ASGI), Uvicorn, Pydantic v2, HTTPX
- Handles incoming requests, data validation, priority scoring, and MongoDB operations.

🧠 AI: Groq Cloud API & Llama 3.3-70B

- Meta Llama 3.3-70B Versatile on Groq LPUs (<1.5s latency, JSON mode)
- Identifies emergencies, extracts hazards, victims & needs; heuristic failover.

⚡ WORKFLOW AUTOMATION — n8n (CENTRAL ORCHESTRATION BACKBONE)

n8n is the central automation and orchestration engine that coordinates connected workflow steps, external channels, and autonomous microservices.

✓ **Webhook Triggers:** Receives real-time intake and lifecycle event payloads from external channels.
✓ **Event Routing & Logic:** Evaluates payload conditions, priority thresholds, and workflow routing rules.
✓ **Service Integration:** Seamlessly connects FastAPI backend services with Groq AI and databases.
✓ **Notification Dispatch:** Coordinates automated dispatches to responder Telegram bots and email queues.
✓ **Human Gate Coupling:** Halts execution until operator approval webhooks are confirmed by commanders.
✓ **Scheduled Monitoring:** Executes background polling loops for incident staleness and liveliness checks.
✓ **Execution Logging:** Records immutable forensic audit traces for every executed workflow node.

💾 Database & Channels: MongoDB & Telegram API

- MongoDB Atlas Cloud with Motor async driver (5 collections) + In-Memory resilience
- Telegram Bot API (intake & markdown alerts), SMTP email triggers, and Leaflet OpenStreetMap.

PUBLIC REPOSITORY URL (CLICKABLE)

https://github.com/jaiwant03/RescueFlow-AI.git

Complete open-source repository containing verified working code, n8n visual automation exports, test suites, and technical documentation.

/frontend

React 18 EOC Console

Production React 18 single-page application with Command Center, live GIS mapping, incident management, approval interface, analytics, and telemetry controls.

/backend

FastAPI Asynchronous Core

FastAPI microservice endpoints, Groq Llama 3.3-70B integration, semantic deduplication, deterministic priority calculation, and MongoDB persistence models.

/n8n

Master Workflow JSON

Production export of n8n master orchestrator workflows (rescueflow_master_orchestrator.json), webhook callbacks, logic routing, and dispatch automation.

/docs & /tests

Technical Docs & Tests

Detailed 16-page technical report, architecture specifications, API documentation, and 20-point verified regression audit suite (test_audit_suite.py).

🌐 Live Production Web Application

Hosted on Vercel High-Performance Global CDN (Clickable):

https://rescue-flow-ai-eight.vercel.app/

Full EOC console with real-time telemetry, interactive Leaflet GIS map, and Approval Center.

🤖 Live Citizen Telegram Bot

Continuous 24/7 Citizen Emergency Intake Channel:

@RescueFlowAI_DemoBot

Search on Telegram and send natural language messages or GPS location pins to test live intake.

Evaluator's 10-Step Triage & Verification Journey

Follow these sequential steps to evaluate the complete end-to-end pipeline in under 2 minutes:

1. Open the deployed RescueFlow AI web application in any browser.
2. Navigate to Emergency Report (/report).
3. Submit a sample emergency report (or use 1-click scenario presets).
4. Observe Groq AI classification and structured data extraction.
5. View incident creation and objective deterministic priority score.
6. Open the Command Center operations dashboard.
7. Observe the incident plotted on the Leaflet GIS map with pulsing marker.
8. Review the incident in the Approval Center (/approvals).
9. Approve the authorized emergency response with coordinator sign-off.
10. Observe the configured n8n notification workflow execute in Response Activity.

OFFICIAL DEMONSTRATION WALKTHROUGH LINK

https://youtu.be/RescueFlowAI-Infinex2k26

⚠ Verification Note: This URL represents the dedicated video demonstration placeholder. Evaluators may also inspect the live Vercel deployment and local video assets.

7-Chapter Demonstration Coverage & Video Script

Chapter 1 — Emergency Intake: Citizen sends live distress report and GPS location via Telegram Bot (@RescueFlowAI_DemoBot) and Web Portal.

Chapter 2 — AI Intelligence: Groq Llama 3.3-70B classifies disaster in ~1.1s, extracts trapped victims, vulnerable groups, and required gear.

Chapter 3 — Intelligent Deduplication: Second citizen sends corroborating flood report; engine matches spatial landmarks and merges into existing ticket.

Chapter 4 — Priority Assessment: Deterministic scoring engine calculates objective 85/100 Critical priority with transparent audit factors.

Chapter 5 — Human-in-the-Loop Approval: Operations Commander inspects incident details, evaluates GIS map, and authorizes response deployment.

Chapter 6 — n8n Automated Dispatch: n8n visual workflow executes dispatches and broadcasts formatted alerts to responder Telegram channels and email queues.

Chapter 7 — Live Command Center: Dashboard reflects updated status, Leaflet GIS map pins, telemetry counters, and immutable compliance audit logs.

“Want to understand the complete architecture, AI intelligence, automation workflows, database design, implementation details and testing of RescueFlow AI?”

Explore the complete technical documentation to discover how RescueFlow AI transforms emergency messages into intelligent, prioritized and coordinated response workflows.

🔗 VIEW COMPLETE PROJECT DOCUMENTATION

Public Document Link Placeholder: [INSERT PUBLIC PROJECT DOCUMENTATION LINK]

Secondary Resource: https://github.com/jaiwant03/RescueFlow-AI.git

Part A: Submission Package

Mandatory official requirements: problem, solution, architecture, GitHub repository, and live deployment touchpoints.

Part B: 19 Technical Sections

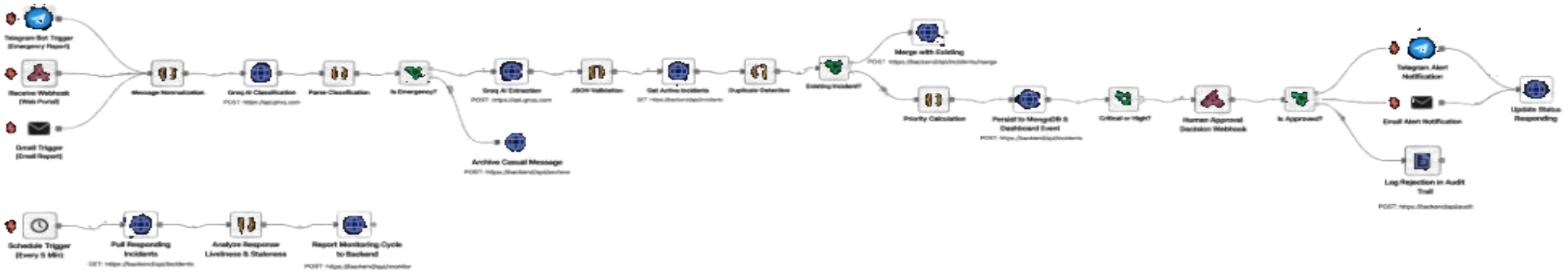
Deep technical audit covering 6 disaster personas, system comparisons, database models, and API endpoints.

10 Verified Test Cases

Empirical validation matrix covering intake, Groq extraction, deduplication, priority scoring, and failover recovery.

Standards & Governance

Sendai Disaster Framework, NDMA Incident Response Guidelines, OASIS CAP-v1.2, and Human-in-the-Loop safety.



“The n8n workflow acts as the automation and orchestration backbone of RescueFlow AI. It connects incoming events, processing services, conditional routing and notification integrations to coordinate the emergency response lifecycle.”

9 VERIFIED WORKFLOW STAGES IMPLEMENTED IN MASTER GRAPH:

1. Emergency Intake • 2. Message Normalization • 3. Groq AI Classification • 4. Structured Data Extraction • 5. Deduplication & Merging
6. Priority Calculation • 7. Human Approval Gate • 8. Automated Multi-Channel Response (Telegram/Email) • 9. Scheduled Response Monitoring